

BTCUSDT Market-Making Project

Backtesting an Avellaneda–Stoikov Strategy on Binance Spot Limit Order Book Data

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Abstract

This report presents the construction of a backtesting framework for a BTCUSDT market-making strategy using Binance Spot limit order book data. The strategy is based on the Avellaneda–Stoikov framework and is enriched with microstructure signals such as bid–ask spread, order book imbalance, top-of-book depth, and short-term volatility. The project includes data loading and cleaning, feature engineering, empirical calibration of fill probabilities as a function of quote distance from the mid-price, execution simulation, inventory management, terminal inventory liquidation, and a detailed PnL decomposition. The main objective is to understand under which conditions a market-making strategy can generate positive gross edge, and how this edge is affected by transaction costs, inventory exposure, and adverse selection.

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1 Introduction

Market making consists of providing liquidity to the market by placing simultaneous limit orders on both the bid and ask sides. A market maker aims to buy at the bid and sell at the ask, thereby capturing part of the bid–ask spread. However, this activity is risky: the market maker can accumulate an unwanted directional position and can be executed precisely when the price is about to move against them. This phenomenon is known as *adverse selection*.

In this project, we build a backtester for a market-making strategy applied to BTCUSDT Spot. The core of the strategy is inspired by the Avellaneda–Stoikov model, which provides a theoretical quoting rule based on the mid-price, volatility, inventory, and risk-aversion parameters. Around this quoting engine, we add more realistic components: microstructure signals, risk filters, fill-probability calibration, transaction costs, and PnL decomposition.

The objective is not to claim that the resulting strategy is immediately tradable in production. The objective is to build a clear research framework that answers the following questions:

- Does the strategy generate positive spread capture?
- Does the PnL come from market making or from directional exposure to Bitcoin?
- How sensitive is the strategy to fees?
- Are the simulated fills favorable, or are they affected by adverse selection?
- Which components should be made more realistic to improve the backtest?

2 Data

The data used in this project consists of Binance Spot limit order book observations for the BTCUSDT symbol. The raw file contains the following columns:

Column	Description
<code>timestamp</code>	Observation date and time
<code>symbol</code>	Traded symbol, here BTCUSDT
<code>best_bid_price</code>	Best bid price
<code>best_bid_qty</code>	Quantity available at the best bid
<code>best_ask_price</code>	Best ask price
<code>best_ask_qty</code>	Quantity available at the best ask
<code>mid_price</code>	Mid-price
<code>spread</code>	Bid–ask spread

In our experiment, the dataset contains 20,000 rows. The first 30% of the observations are used for calibration, while the remaining 70% are used for out-of-sample backtesting. Therefore:

- 6,000 observations are used for calibration;
- 14,000 observations are used for backtesting.

The tick size used in the backtest is 0.01 USDT, which is consistent with the observed price grid in the BTCUSDT Spot data.

3 Microstructure Features

Several microstructure variables are computed from the order book data. These variables are used to determine whether the market is suitable for market making and to adjust quotes according to market conditions.

3.1 Mid-Price

The mid-price is defined as:

$$m_t = \frac{b_t^{\text{best}} + a_t^{\text{best}}}{2}, \quad (1)$$

where b_t^{best} is the best bid and a_t^{best} is the best ask.

The mid-price is the central reference price. The strategy places quotes around this level, with adjustments based on inventory and microstructure signals.

3.2 Spread and Spread in Basis Points

The absolute spread is:

$$\text{spread}_t = a_t^{\text{best}} - b_t^{\text{best}}. \quad (2)$$

To compare spreads across different price levels, we also compute the spread in basis points:

$$\text{spread_bps}_t = 10,000 \times \frac{\text{spread}_t}{m_t}. \quad (3)$$

The spread in basis points measures the relative cost of crossing the book. In our dataset, the spread is extremely tight: the median spread is around 0.0013 bps. This means that BTCUSDT is very liquid at the top of the book, but it also makes market making difficult when fees are significant.

3.3 Bid/Ask Imbalance

The top-of-book imbalance is defined as:

$$I_t = \frac{Q_t^{\text{bid}} - Q_t^{\text{ask}}}{Q_t^{\text{bid}} + Q_t^{\text{ask}}}, \quad (4)$$

where Q_t^{bid} is the quantity at the best bid and Q_t^{ask} is the quantity at the best ask.

This quantity lies in $[-1, 1]$:

- $I_t > 0$ means that bid-side quantity is larger than ask-side quantity;
- $I_t < 0$ means that ask-side quantity is larger than bid-side quantity;
- $I_t \approx 0$ means that the top of the book is balanced.

The intuition is the following: if the bid-side quantity is much larger than the ask-side quantity, the order book may reflect buying pressure, and the price may move upward in the short term. In that case, selling too aggressively at the ask is dangerous. Symmetrically, if the ask side dominates, buying too aggressively at the bid can expose the strategy to a short-term price decrease.

3.4 Order Book Depth

The top-of-book depth is defined as:

$$D_t = Q_t^{\text{bid}} + Q_t^{\text{ask}}. \quad (5)$$

Low depth means that the book is thin and that the price may move quickly after an aggressive order. The strategy therefore avoids quoting when top-of-book depth is too low.

3.5 Returns and Short-Term Volatility

We compute log-returns of the mid-price:

$$r_t = \log(m_t) - \log(m_{t-1}). \quad (6)$$

Short-term volatility is estimated as the rolling standard deviation of these returns:

$$\sigma_t = \text{Std}(r_{t-w+1}, \dots, r_t), \quad (7)$$

where w is the rolling window size, set to 100 observations in our experiments.

Volatility is a central input in the Avellaneda–Stoikov framework: when volatility increases, the risk of carrying inventory increases, and quotes should become more conservative.

4 The Avellaneda–Stoikov Model

The Avellaneda–Stoikov model provides a classical framework for optimal market making. The idea is to compute a reservation price and then place a bid and an ask around that price.

4.1 Reservation Price

The reservation price is given by:

$$r_t = m_t - q_t \gamma \sigma_t^2 T, \quad (8)$$

where:

- m_t is the mid-price;
- q_t is the current inventory;
- γ is the risk-aversion coefficient;
- σ_t is the short-term volatility estimate;
- T is the effective horizon used in the formula.

The intuition is straightforward. If the strategy is long, meaning $q_t > 0$, the reservation price decreases. This makes the strategy more willing to sell and less willing to buy. If the strategy is short, meaning $q_t < 0$, the reservation price increases, which makes the strategy more willing to buy back inventory.

Thus, the model naturally introduces inventory control.

4.2 Optimal Spread

The theoretical total spread is approximated by:

$$\delta_t = \gamma \sigma_t^2 T + \frac{2}{\gamma} \log \left(1 + \frac{\gamma}{\kappa} \right), \quad (9)$$

where κ controls how fast the execution intensity decreases as quotes move farther from the mid-price.

The first part of the spread,

$$\gamma \sigma_t^2 T, \quad (10)$$

corresponds to inventory risk. The higher the volatility, the wider the optimal spread should be.

The second part,

$$\frac{2}{\gamma} \log \left(1 + \frac{\gamma}{\kappa} \right), \quad (11)$$

captures the trade-off between execution probability and profit per trade. Quoting close to the mid leads to more executions but smaller margins. Quoting farther away leads to fewer executions but higher profit per fill.

4.3 Bid and Ask Quotes

The raw quotes are computed as:

$$b_t = r_t - \frac{\delta_t}{2}, \quad a_t = r_t + \frac{\delta_t}{2}. \quad (12)$$

These quotes are then adjusted using microstructure signals and rounded to the tick size.

5 Inventory Skew and Imbalance Skew

5.1 Inventory Control

Inventory is one of the main risks faced by a market maker. If the strategy buys more often than it sells, it becomes long BTC. If the price then decreases, the strategy suffers a mark-to-market loss.

The Avellaneda–Stoikov model already partially addresses this issue through the reservation price. In the backtester, we also add constraints such as:

- if inventory exceeds an upper limit, the strategy stops placing bids;
- if inventory is too negative, the strategy stops placing asks;
- at the end of the backtest, any remaining inventory is liquidated.

Terminal liquidation is important: without it, a strategy may appear profitable simply because it ends the backtest with an unrealized position valued favorably at the mid-price.

5.2 Imbalance Skew

Imbalance is used to adjust quotes according to short-term pressure observed in the book. We apply a skew of the form:

$$\text{skew}_t = \alpha_I I_t \cdot \text{spread}_t, \quad (13)$$

where α_I controls the strength of the adjustment.

The quotes are then shifted as follows:

$$b_t \leftarrow b_t + \text{skew}_t, \quad a_t \leftarrow a_t + \text{skew}_t. \quad (14)$$

If $I_t > 0$, the bid side is stronger than the ask side. This suggests potential upward pressure. Quotes are shifted upward: the bid becomes more aggressive and the ask becomes less aggressive, reducing the risk of selling too cheaply before an upward move.

If $I_t < 0$, quotes are shifted downward: the ask becomes more aggressive and the bid becomes less aggressive, reducing the risk of buying too high before a downward move.

The spread is used as a price scale: imbalance is dimensionless, whereas a quote skew must be expressed in price units.

6 Fill Probability Calibration

A key part of the project is the empirical calibration of execution probabilities. Choosing fill probabilities arbitrarily would make the backtest unreliable.

6.1 Principle

At each time t , we construct virtual quotes at different distances from the mid-price. For a distance of d ticks:

$$b_t^{(d)} = m_t - d \cdot \text{tick_size}, \quad a_t^{(d)} = m_t + d \cdot \text{tick_size}. \quad (15)$$

We then check whether the mid-price touches these levels over the next h observations.

For the bid, we compute:

$$\min_{1 \leq u \leq h} m_{t+u}. \quad (16)$$

The virtual bid is considered touched if:

$$\min_{1 \leq u \leq h} m_{t+u} \leq b_t^{(d)}. \quad (17)$$

For the ask, we compute:

$$\max_{1 \leq u \leq h} m_{t+u}, \quad (18)$$

and the virtual ask is considered touched if:

$$\max_{1 \leq u \leq h} m_{t+u} \geq a_t^{(d)}. \quad (19)$$

This gives an approximation of the probability that an order placed d ticks away from the mid-price is executed within horizon h .

6.2 Exponential Fill Model

We then fit a model of the form:

$$p_{fill}(d) \approx p_0 e^{-k_{fill} d}, \quad (20)$$

where:

- p_0 is the base probability of being executed near the mid;
- k_{fill} controls how fast the fill probability decreases with distance;
- d is the distance in ticks.

Taking logarithms gives:

$$\log p_{fill}(d) \approx \log p_0 - k_{fill} d. \quad (21)$$

Therefore, p_0 and k_{fill} can be estimated through a linear regression of $\log p_{fill}(d)$ on d .

In our experiment, the calibration gives approximately:

Parameter	Calibrated value
p_0	0.1745
k_{fill}	0.00181
κ	0.1813

The relation between k_{fill} and κ comes from the fact that k_{fill} is expressed per tick, while κ is expressed per price unit. Since:

$$\delta_{price} = d \cdot \text{tick_size}, \quad (22)$$

we obtain:

$$\kappa = \frac{k_{fill}}{\text{tick_size}}. \quad (23)$$

6.3 Limitations of This Calibration

This calibration is an approximation. It measures whether the price level was touched, but it does not guarantee that a real order would have been executed. In a real limit order book, execution also depends on:

- queue position;
- volume ahead of our order;
- market order flow;
- cancellations;
- latency.

Nevertheless, this approximation is useful as a first step because it makes the fill model data-driven.

7 Execution Model

In the first version of the backtester, fills are simulated probabilistically. For a quote at distance d ticks from the mid-price, we use:

$$p_{fill}(d) = p_0 e^{-k_{fill} d}. \quad (24)$$

At each timestamp, if the strategy places a bid or an ask, a uniform random variable is drawn. If this random variable is below the fill probability, the order is considered executed.

The sign conventions are:

- a fill at the bid is a buy, hence a positive signed quantity;
- a fill at the ask is a sell, hence a negative signed quantity.

If q_i is the signed quantity of trade i and p_i is its execution price, the impact on cash is:

$$\Delta \text{cash}_i = -q_i p_i - \text{fee}_i. \quad (25)$$

This formula works for both buys and sells. If $q_i > 0$, the strategy buys and cash decreases. If $q_i < 0$, the strategy sells and cash increases.

8 Inventory Management and Terminal Liquidation

Inventory evolves according to:

$$Q_t = \sum_{i \leq t} q_i. \quad (26)$$

The mark-to-market PnL is:

$$\text{PnL}_t = \text{cash}_t + Q_t m_t. \quad (27)$$

At the end of the backtest, remaining inventory is liquidated:

- if $Q_T > 0$, the inventory is sold at the best bid;
- if $Q_T < 0$, the inventory is bought back at the best ask.

This liquidation avoids overestimating performance by valuing an unrealized final position at the mid-price. After liquidation, final inventory is zero and the PnL is fully realized.

9 PnL Decomposition

PnL decomposition is essential for understanding what the strategy is actually doing. We distinguish several components.

9.1 Total PnL

Total PnL is:

$$\text{PnL}_T = \text{cash}_T + Q_T m_T. \quad (28)$$

After terminal liquidation, $Q_T = 0$, hence:

$$\text{PnL}_T = \text{cash}_T. \quad (29)$$

9.2 Spread Capture

Spread capture measures the immediate gain obtained from buying below the mid-price or selling above the mid-price. For a buy:

$$\text{SC}_i = q_i(m_i - p_i), \quad q_i > 0. \quad (30)$$

For a sell:

$$\text{SC}_i = |q_i|(p_i - m_i), \quad q_i < 0. \quad (31)$$

Positive spread capture indicates that the strategy buys below the mid and sells above the mid, which is the desired behavior for a market maker.

9.3 Inventory PnL

Inventory PnL measures the PnL generated by carrying a position while the mid-price moves:

$$\text{InventoryPnL} = \sum_t Q_{t-1}(m_t - m_{t-1}). \quad (32)$$

This component is important because it helps distinguish genuine market-making PnL from directional PnL. If a large part of the PnL comes from inventory, the strategy is exposed to market direction.

9.4 Fees

Fees are computed trade by trade:

$$\text{fee}_i = f|q_i|p_i, \quad (33)$$

where f is the fee rate.

Total fees are:

$$\text{Fees} = \sum_i f|q_i|p_i. \quad (34)$$

Fees are crucial in this project because the BTCUSDT spread is extremely tight. A strategy that appears profitable without fees may become unprofitable once realistic fees are included.

9.5 Adverse Selection

Adverse selection measures whether the price moves against the strategy after execution. For a horizon h , we compute:

$$AS_i = q_i(m_{i+h} - m_i). \quad (35)$$

For a buy, if the mid-price increases after the fill, this quantity is positive: the fill was favorable. If the mid-price decreases, it is negative: the strategy bought before a price drop.

For a sell, $q_i < 0$. If the mid-price decreases after the sale, the quantity is positive. If the mid-price increases, it is negative.

Thus, a negative average adverse-selection metric indicates that the strategy is being executed at unfavorable times.

10 Code Architecture

The project is organized modularly:

File	Role
<code>data_loader.py</code>	Data loading, validation, and cleaning
<code>features.py</code>	Microstructure feature engineering
<code>calibration.py</code>	Fill-probability and risk-threshold calibration
<code>quote_engine.py</code>	Avellaneda–Stoikov quotes and imbalance skew
<code>risk.py</code>	Risk filters and quote permissions
<code>execution.py</code>	Probabilistic fill simulation
<code>strategy.py</code>	Strategy logic, cash, inventory, and PnL
<code>backtester.py</code>	Backtest loop over all observations
<code>pnl.py</code>	PnL decomposition
<code>metrics.py</code>	Performance metrics
<code>run_backtest.py</code>	Main backtest script

This separation is important because it makes the project easier to read, test, and improve. For example, the execution model can be replaced without changing the quote engine, and the feature set can be improved without modifying the PnL logic.

11 Experimental Results

The strategy is backtested on 14,000 out-of-sample observations. The base order size is set to 0.001 BTC and the maximum inventory is set to 0.01 BTC.

11.1 Zero-Fee Result

In the zero-fee case, the strategy produces the following results:

Metric	Value
Number of trades	955
Final PnL	14.68 USDT
Spread capture	4.19 USDT
Inventory PnL	10.49 USDT
Final inventory	0 BTC
Max drawdown	−1.73 USDT

The PnL is positive, but a significant part of it comes from inventory PnL. This means that the strategy benefited from favorable directional BTC exposure during the backtest.

11.2 Fee Sensitivity

We then perform a fee-sensitivity analysis. The results are:

Fee rate	Total fees	Final PnL	Non-annualized Sharpe
0.00000	0.0000	14.6801	0.0384
0.00001	0.7378	13.9423	0.0365
0.00005	3.6892	10.9909	0.0288
0.00010	7.3785	7.3016	0.0191
0.00020	14.7570	-0.0769	-0.0002

These results show that the strategy has positive gross edge in the simplified backtest, but this edge is almost entirely consumed when the fee rate reaches 0.0002, i.e. approximately 2 bps per trade.

The break-even fee rate can be estimated as:

$$f_{BE} = \frac{\text{GrossPnL}}{\sum_i |q_i| p_i}. \quad (36)$$

Using the run with $f = 0.0002$, we get approximately:

$$f_{BE} \approx 0.0002 \times \frac{14.6801}{14.7570} \approx 0.000199. \quad (37)$$

Therefore, the strategy is close to break-even for fees around 2 bps per trade.

12 Economic Interpretation

The results highlight several important points.

First, the strategy does capture spread: spread capture is positive. This means that the quotes are placed coherently around the mid-price.

Second, a large part of the PnL comes from inventory PnL. This means that the strategy is not a perfectly neutral market maker: it sometimes carries directional exposure. This is not necessarily bad, but it must be clearly identified. A strategy that profits mainly from inventory exposure is closer to a directional strategy than to pure market making.

Third, profitability is highly sensitive to fees. On BTCUSDT Spot, the top-of-book spread is extremely tight. Therefore, even a small error in the fill model or in the fee assumption can completely change the conclusion.

Fourth, adverse selection is a central metric. If fills tend to be followed by unfavorable mid-price moves, the strategy is providing liquidity at the wrong times.

13 Backtest Limitations

The current backtest is a first version and is intentionally simplified. The main limitations are the following.

13.1 Simplified Fill Model

The execution model is probabilistic and depends only on quote distance from the mid-price. It does not account for:

- queue position;
- actual volume consumed at the bid and ask;

- cancellations;
- market order flow;
- latency.

A more realistic model should simulate the position of the order in the queue and determine whether aggressive flow is sufficient to reach it.

13.2 Liquidity Assumption

The backtest assumes that the order is executed at the simulated quote with the desired quantity. In reality, partial execution is possible.

13.3 Fees and Rebates

Fees are modeled as a simple fixed percentage. In practice, fees depend on VIP level, maker/taker status, and may include rebates. For a real market maker, the fee structure is crucial.

13.4 Limited Test Period

The dataset is relatively short. A more robust validation should use multiple days or weeks and include different market regimes: calm, volatile, directional, and mean-reverting.

14 Possible Improvements

Several natural improvements can be made to the project.

14.1 Queue-Based Execution Model

The most important improvement would be to replace the probabilistic execution model with a queue-based model. The idea would be to track the volume ahead of our order at a given price level and simulate execution only if aggressive orders consume that volume.

14.2 Adverse-Selection Filter

A toxicity score could be added using:

- imbalance;
- short-term volatility;
- rapid mid-price changes;
- order book depth;
- large spread movements.

The strategy could stop quoting or widen quotes when the toxicity score is too high.

14.3 Parameter Optimization

Parameters such as γ , inventory limits, imbalance-skew intensity, and order size could be optimized on a calibration period and then validated out of sample.

14.4 Stricter Inventory Control

Since a large part of the PnL comes from inventory PnL, it would be useful to test stricter inventory controls:

- reducing the inventory limit;
- applying stronger skew when inventory approaches the limit;
- partially liquidating inventory when it remains on the same side for too long.

14.5 Multi-Period Backtesting

The strategy should be tested on several disjoint periods to verify robustness and avoid overfitting to a single market regime.

15 Conclusion

This project developed a complete backtesting framework for an Avellaneda–Stoikov market-making strategy applied to BTCUSDT Spot. The framework includes order book data loading, microstructure feature computation, empirical fill-probability calibration, inventory-aware quote computation, execution simulation, cash and inventory management, terminal liquidation, and PnL decomposition.

The results show that the strategy generates positive gross edge in a simplified environment. However, this edge is highly sensitive to transaction costs. With a fee rate close to 0.0002, the strategy becomes close to break-even. In addition, a significant part of the PnL comes from inventory exposure, showing that the strategy is not yet a pure neutral market maker.

The main conclusion is that market-making profitability does not depend only on the quoting formula. It depends crucially on execution quality, fees, adverse selection, inventory management, and the ability to avoid toxic market regimes. The developed framework therefore provides a solid foundation for moving toward a more realistic backtest incorporating queue position, market order flow, and more advanced toxicity filters.